

WASP-135b

R-0209

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-135_230807 · created 2026-07-16T23:28:47+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460163.8258 +/- 0.0097 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.173 +/- 0.05
Transit depth (Rp/Rs)^2	3.0 +/- 1.72 [%]
Orbital Inclination (inc)	80.91 +/- 2.78
Airmass coefficient 1 (a1)	0.976 +/- 0.026
Airmass coefficient 2 (a2)	0.0111 +/- 0.008
Scatter in the residuals of the lightcurve fit is	2.69 %
Best Comparison Star	#1 - [33, 295]
Optimal Aperture	3.42
Optimal Annulus	14.63
Transit Duration (day)	0.034 +/- 0.018

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.173 ± 0.05 vs 0.139 (deviation 0.63σ)
Duration fit vs published	0.034 d vs 0.0687 d (deviation 1.78σ)
Mid-time O–C	-25.3 min
Depth SNR	3.2
Residual scatter	2.69 %

Light curve

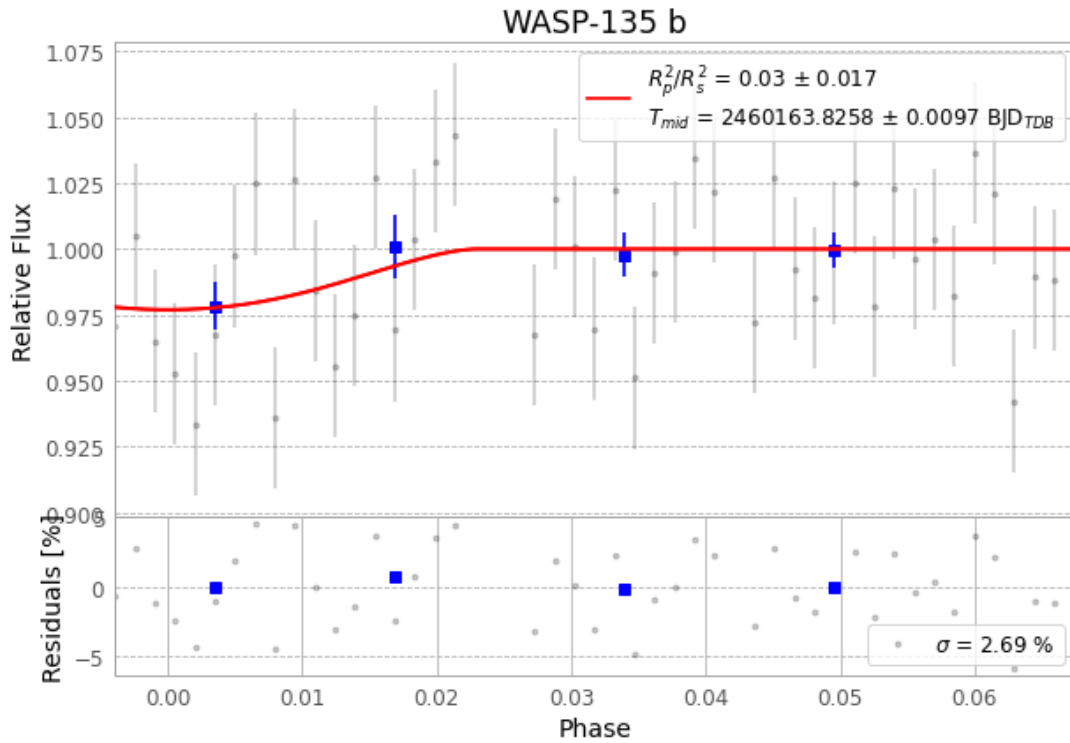


Figure 1 — FinalLightCurve_WASP-135 b_2023-08-07.png (SHA-256 6ad5a4fe6fc9797c...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [263, 162], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 b71e4d06d1e793ee...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1965f5e

Data provenance

48 FITS frames (32 MB), WASP-135230807073615.FITS → WASP-135230807100015.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-135-230807.log (0ee5578f7d6b...), FinalLightCurve_WASP-135 b_2023-08-07.png (6ad5a4fe6fc9...), FinalLightCurve_WASP-135 b_2023-08-07.csv (b33dfc66ad5a...)

Compute resources

Wall time 95.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-16 23:28Z.